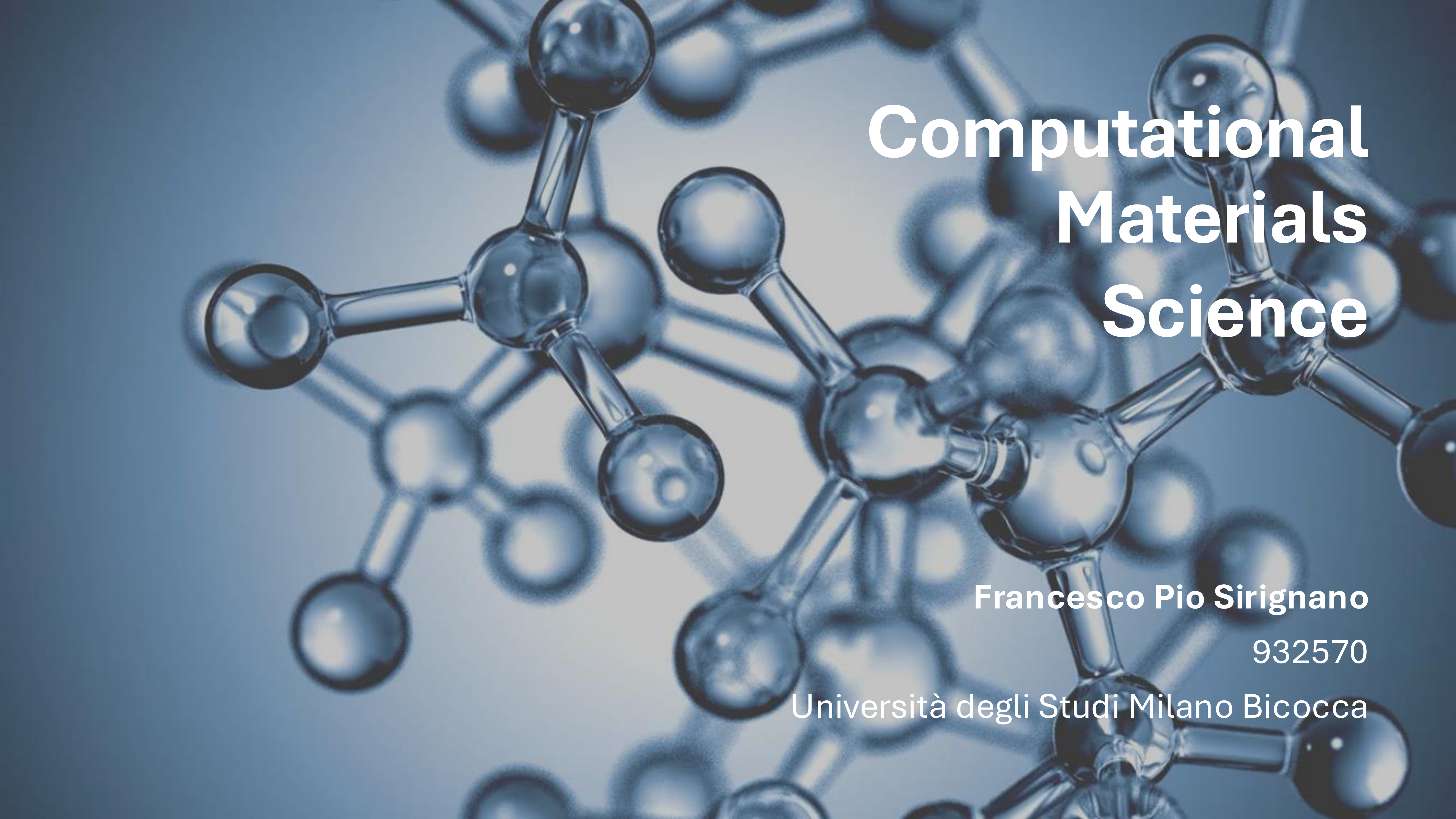

Computational Materials Science

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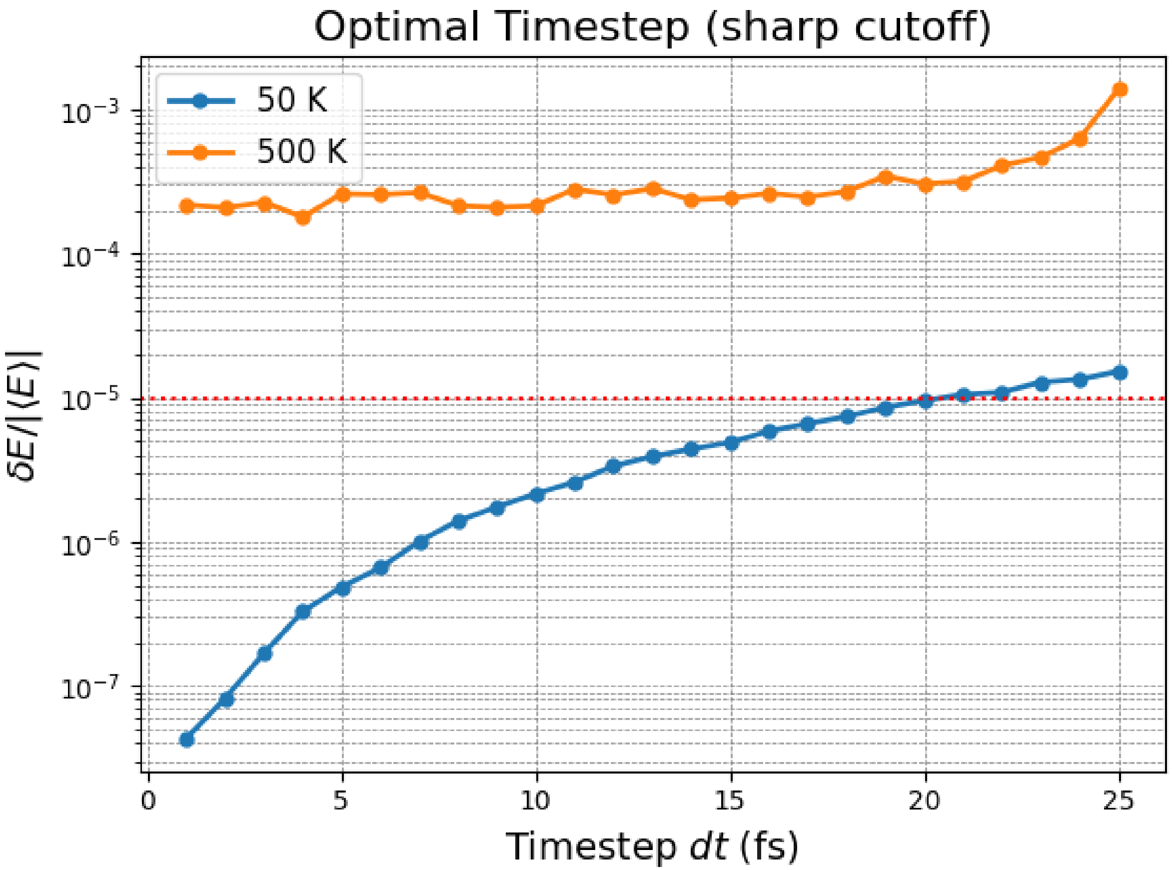
Exercise 1: MD simulations of LJ - Ag

1. Optimal timestep: Sharp cutoff

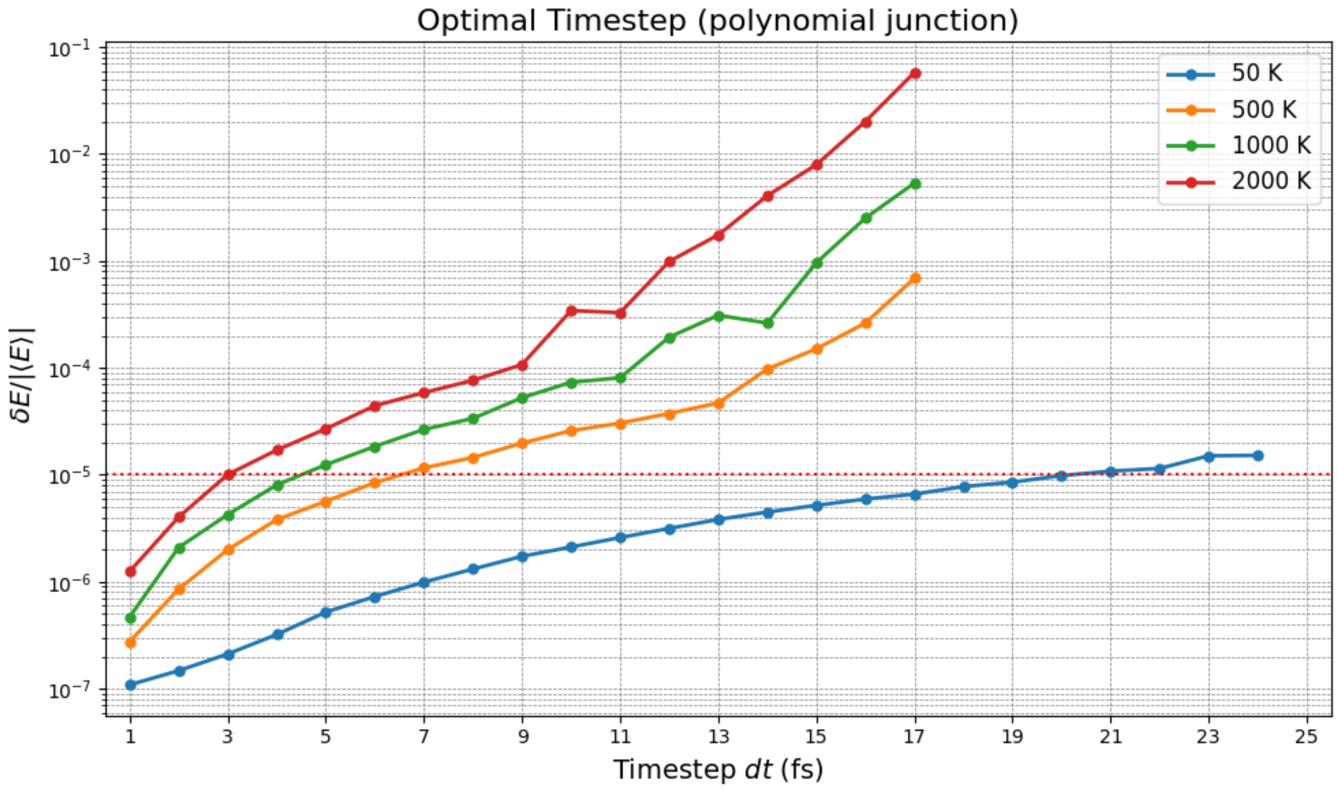


$\boldsymbol{\varepsilon}$ $(\boldsymbol{eV})$	0.345
$\boldsymbol{\sigma}$ (Å)	2.644
$\boldsymbol{r_C}$ (Å)	4.5
$\boldsymbol{\delta_E/\langle E\rangle}$	1e-5

$\boldsymbol{T}$ $(\boldsymbol{K})$	$\boldsymbol{dt}$ $(\boldsymbol{fs})$
50	19
500	-

No optimal timestep at T = 500K, the curve is always > 1e-5

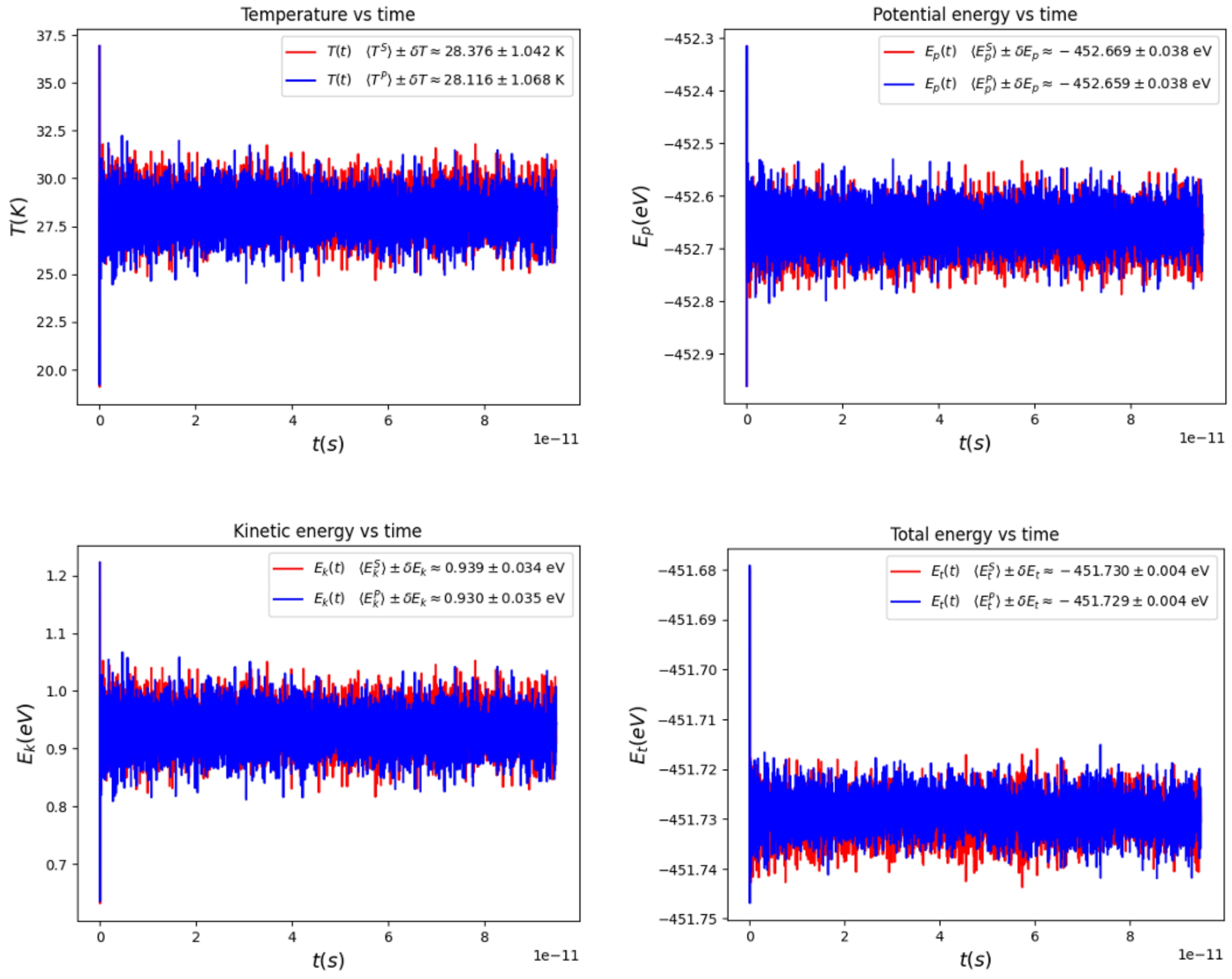
2. Optimal timestep: Polynomial junction



$\varepsilon\ (eV)$	0.345
σ (Å)	2.644
r_C (Å)	4.5
r_P (Å)	4.2
$\delta_E/\langle E\rangle$	1 e-5

$T\ (K)$	$dt\ (fs)$
50	19
500	6
1000	4
2000	2

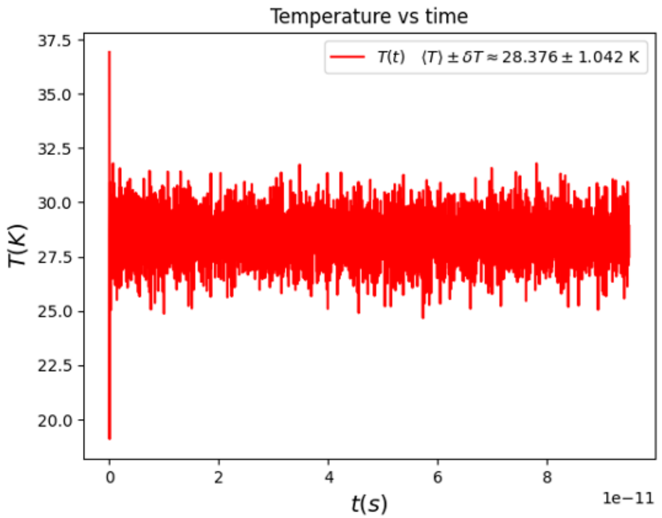
2. Sharp cutoff vs Polynomial junction at 0 K



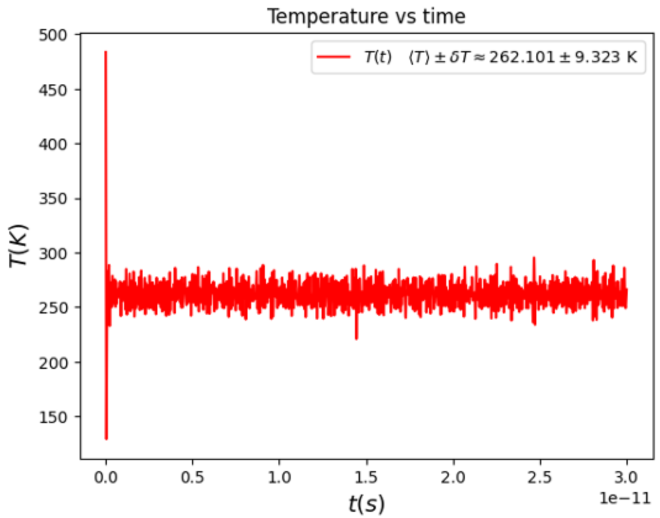
ε (eV)	0.345
σ (Å)	2.644
r_C (Å)	4.5
r_P (Å)	4.2
T (K)	50
dt (fs)	19
t_{th} (ps)	3
Steps	5000

- The simulations at 50 K using the polynomial junction and the sharp-cutoff are consistent, because T is too low to effectively see the effects of the polynomial junction

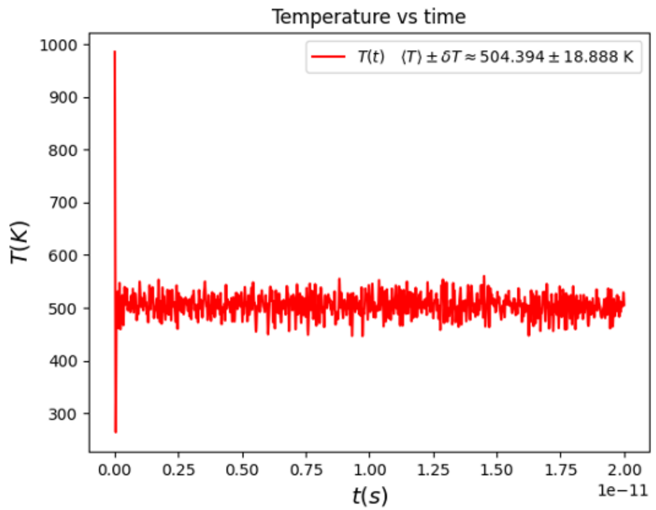
2. Polynomial junction - Temperature



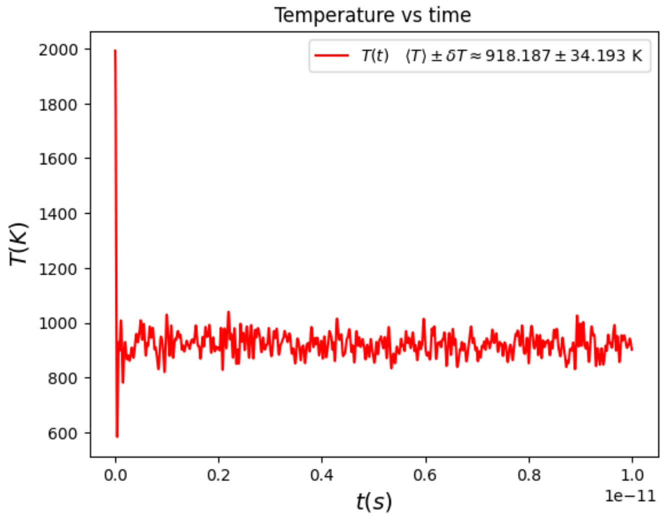
50 K



500 K



1000 K

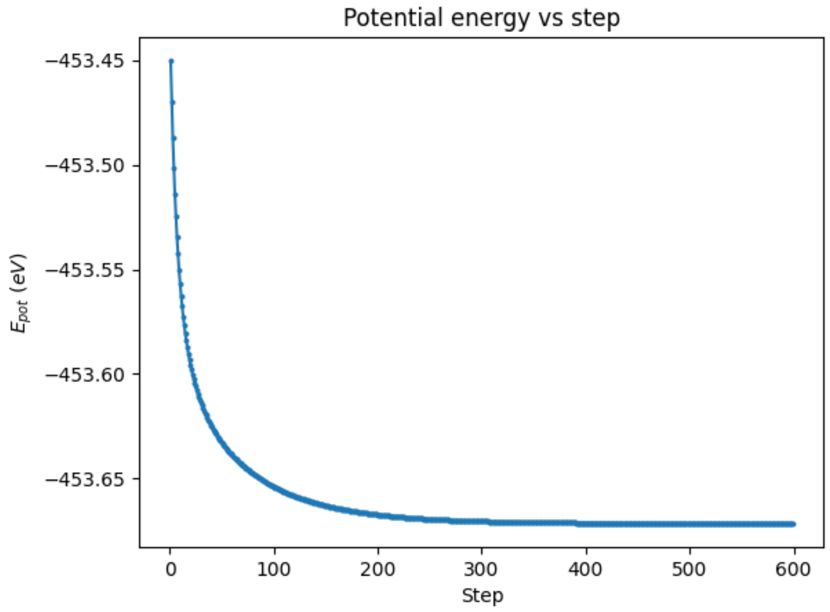


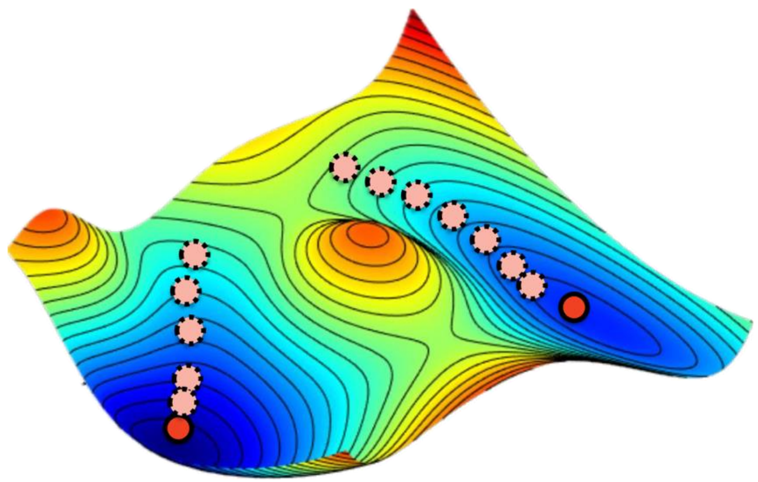
2000 K

$\varepsilon\ (eV)$	0.345
σ (Å)	2.644
r_C (Å)	4.5
r_P (Å)	4.2
$t_{th}\ (ps)$	3
Steps	5000

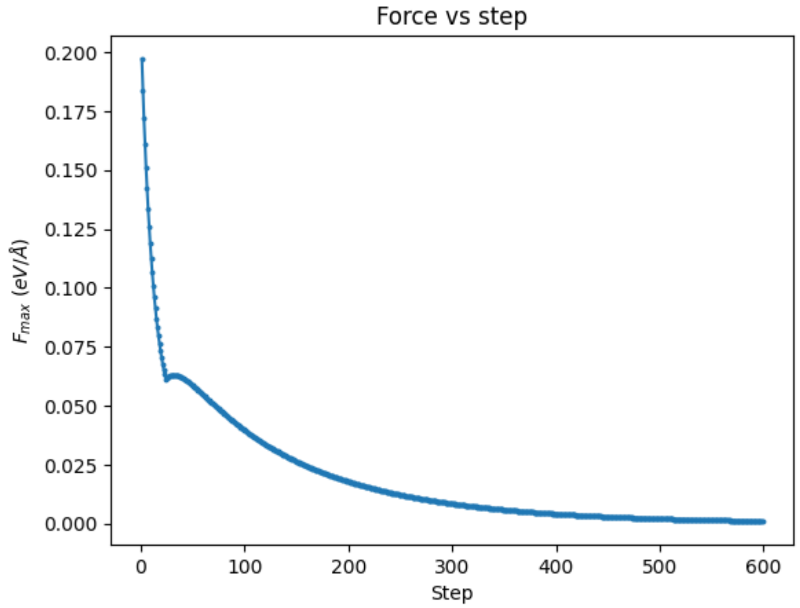
- The condition $\langle T \rangle = \frac{T_{ini}}{2}$ is still not fulfilled, since we have yet to implement the Steepest Descent procedure (we don't start in a minimum)

3. Steepest descent - 10 ps simulations



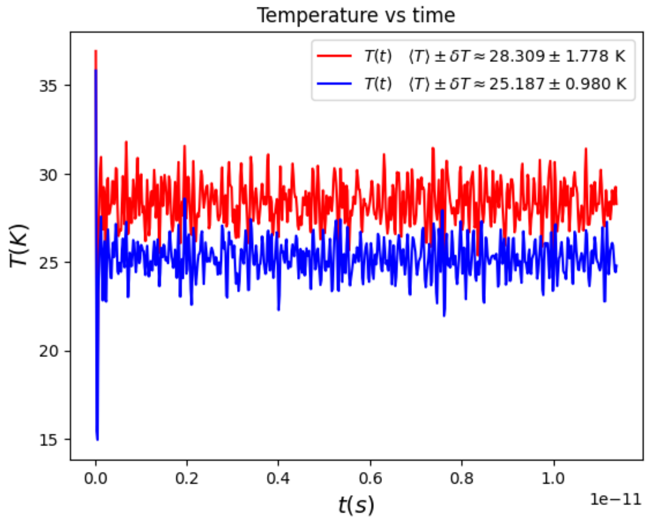


C (Å/eV)	0.005
F_{toll} (eV/Å)	0.001
dt (fs)	19
t_{tot} (ps)	10

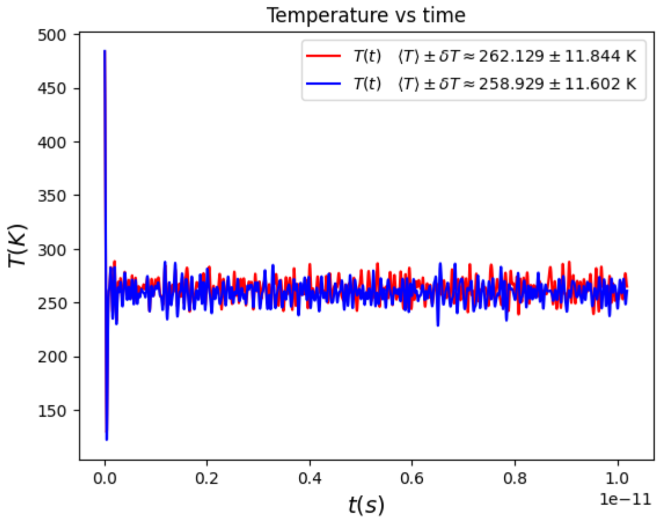


- Energy is monotonously decreasing as required for a minimization algorithm
- A local increment in the force is admissible since we are minimizing energy

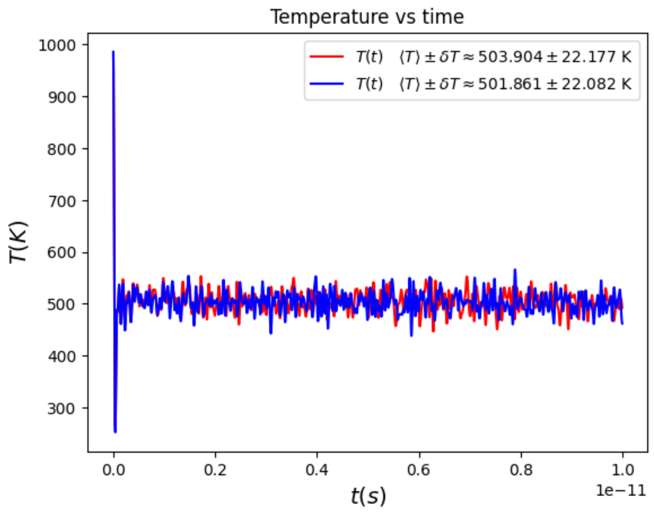
3. Steepest descent - 10 ps simulations



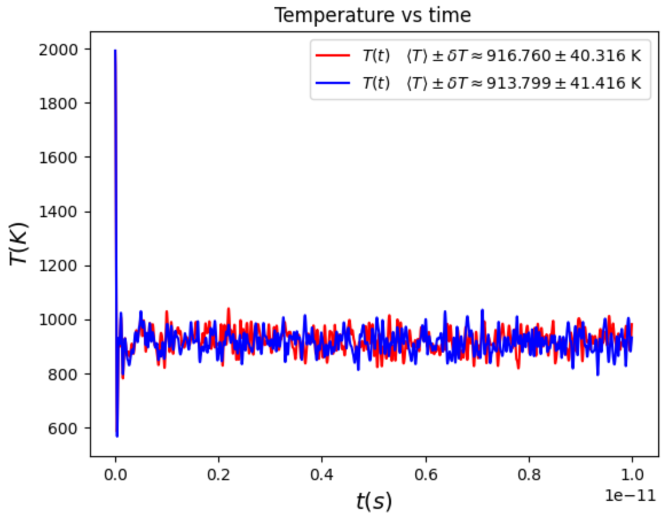
50 K



500 K



1000 K



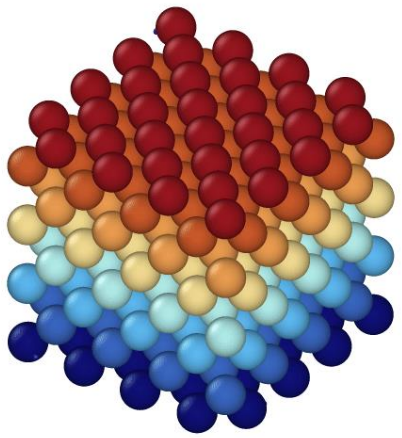
2000 K

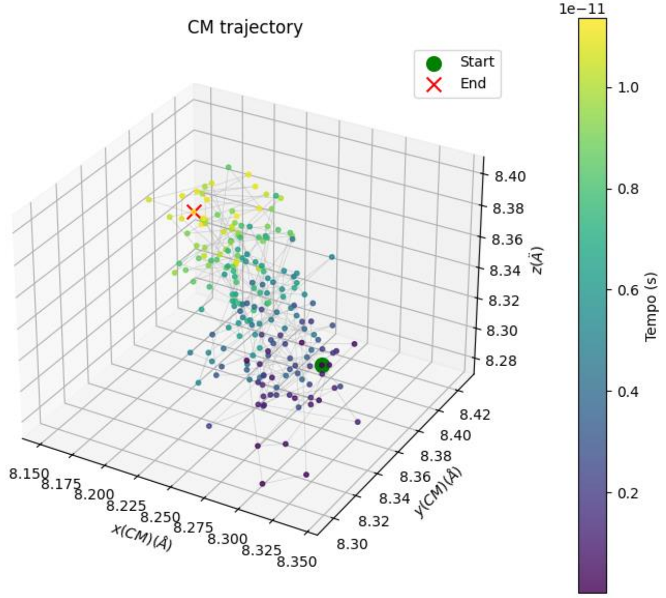
C (Å/eV)	0.005
F_{toll} (eV/Å)	0.001
$dt\ (fs)$	19
$t_{tot}\ (ps)$	10

- The condition $\langle T \rangle = \frac{T_{ini}}{2}$ is now fulfilled, because thanks to the Steepest Descent procedure we start from a local minimum of potential energy.
- LJ works well for small displacements, while at 2000K the atoms move too much → $\langle T \rangle \neq \frac{T_{ini}}{2}$

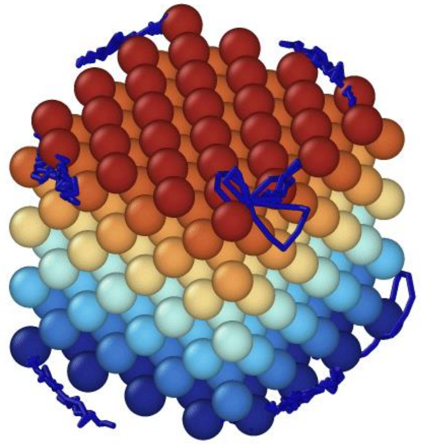
3. Steepest descent - 10 ps simulations

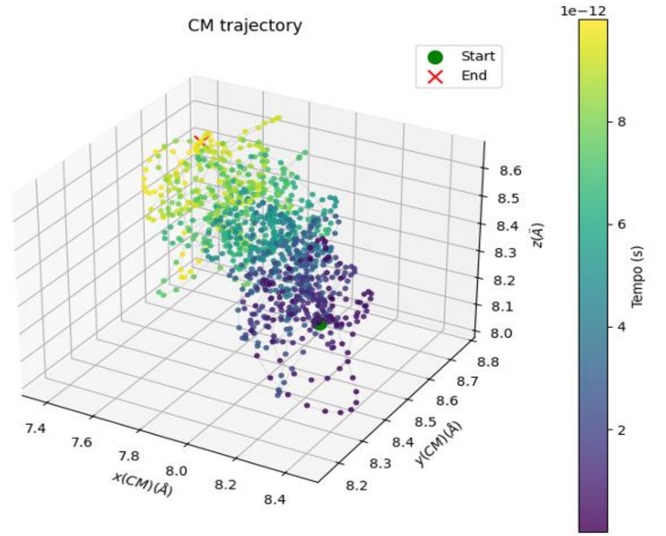
50 K





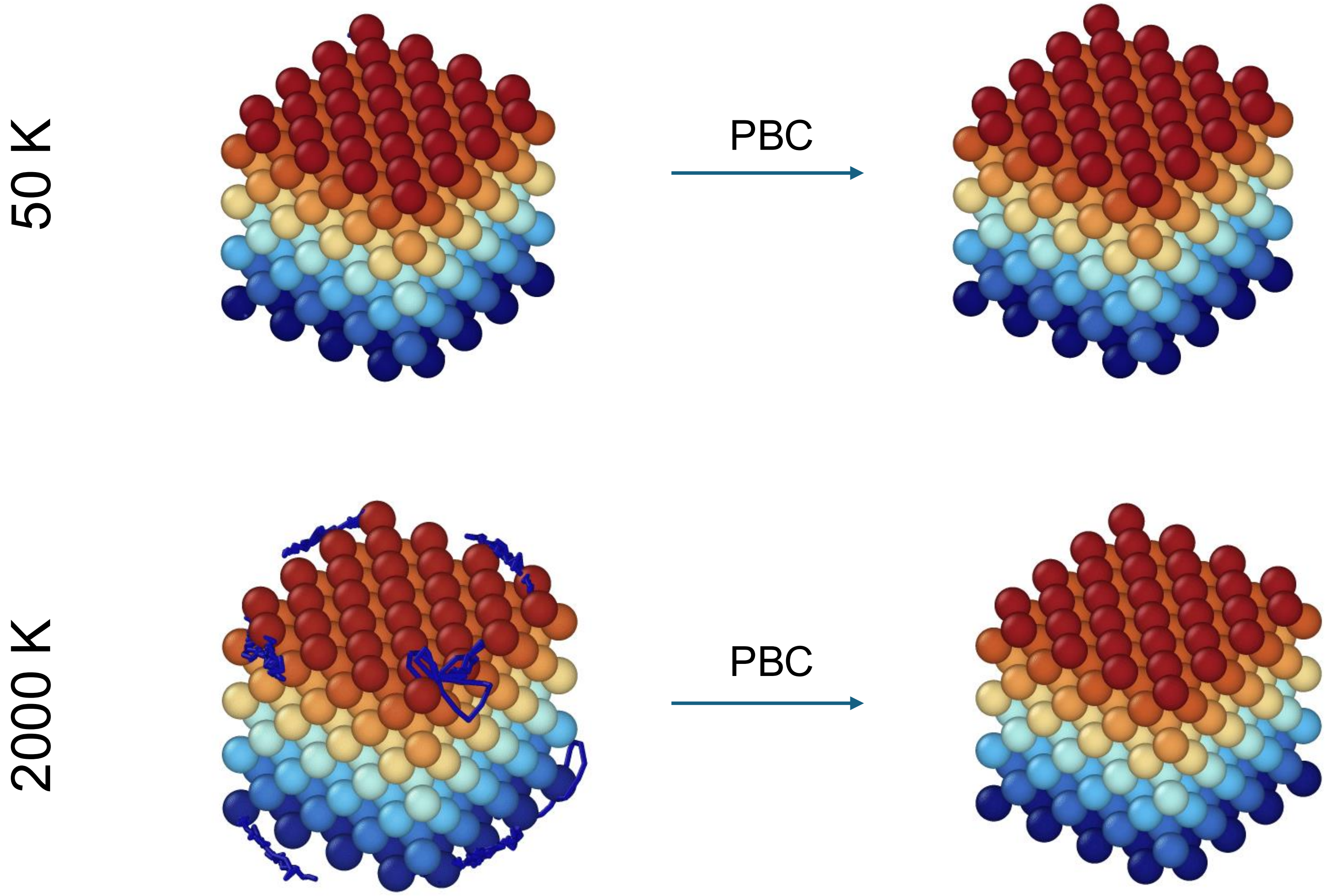
2000 K





- At 2000K the atoms vibrate much more
- From the trajectories at the vertices and the CM we see how the cluster rotates in 3D

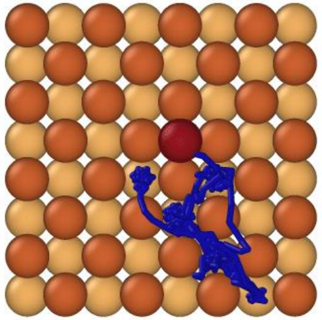
3. Steepest descent - 10 ps simulations



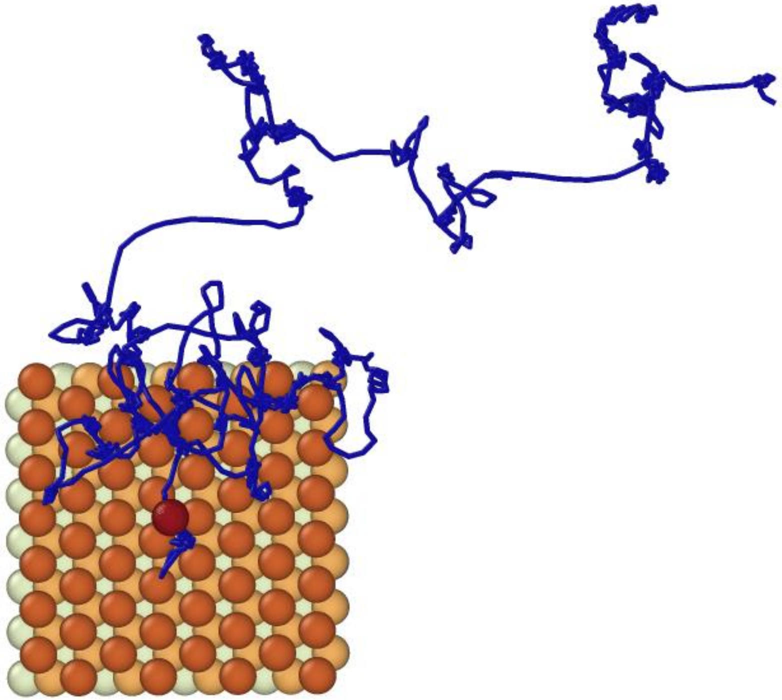
- At 2000K the atoms vibrate much more
- From the trajectories at the vertices and the CM we see how the cluster rotates in 3D
- Implementing PBC stabilizes the movements
  → we can add an adatom

4-5. PBC and simulations with adatom

Fcc (100)



Fcc (111)



T_{ini} (K)	1700
$dt\ (fs)$	2
Tot steps	150000
Sim time (ps)	300
L_x , L_y (Å)	16.6416 (100) 20.3817 (111)

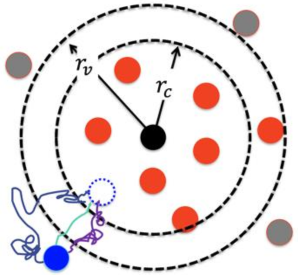
- Fcc(100) more compact than Fcc(111), the adatom is in a deeper well (4 nn for (100) and 3 nn for (111)) → Higher barrier to move to another site

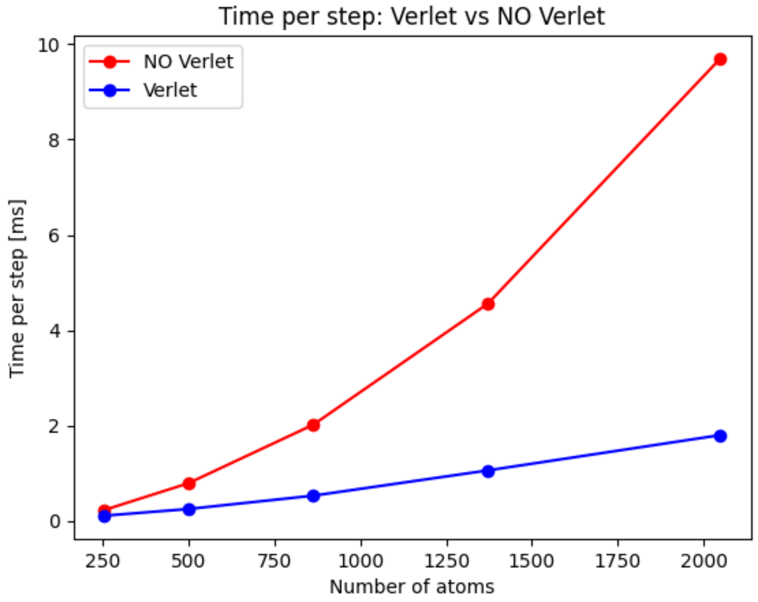
- Diffusion unlikely to happen at room temperature in 300 ps for (100), it's too rare

- Diffusion more probable for (111)

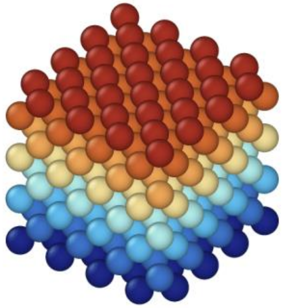
Verlet Cages: difference in time of execution

- **Verlet cage** = 'attention region' of radius $r_V > r_C$ around each atom
- Update the neighbors list only if at least one atom moves more than $\frac{1}{2}(r_V - r_C)$
- NxN scaling → N scaling

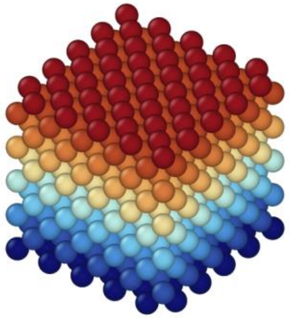




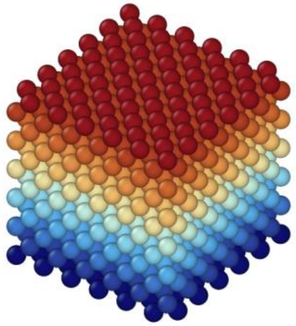
natoms	Before	After
256	0.23	0.11
500	0.79	0.25
864	2.02	0.53
1372	4.56	1.06
2048	9.7	1.8



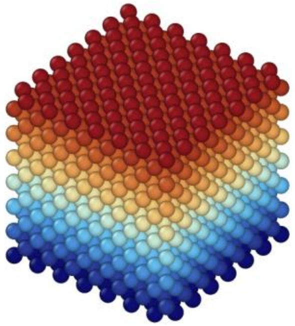
256



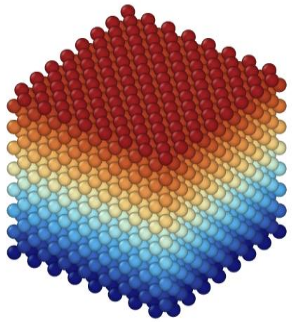
500



864



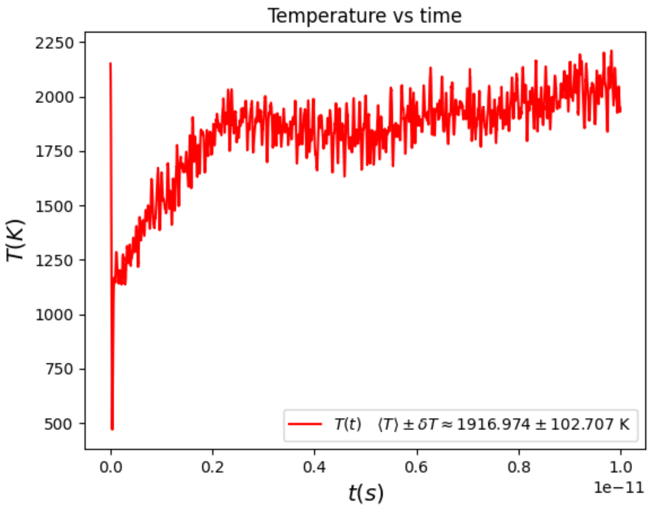
1372

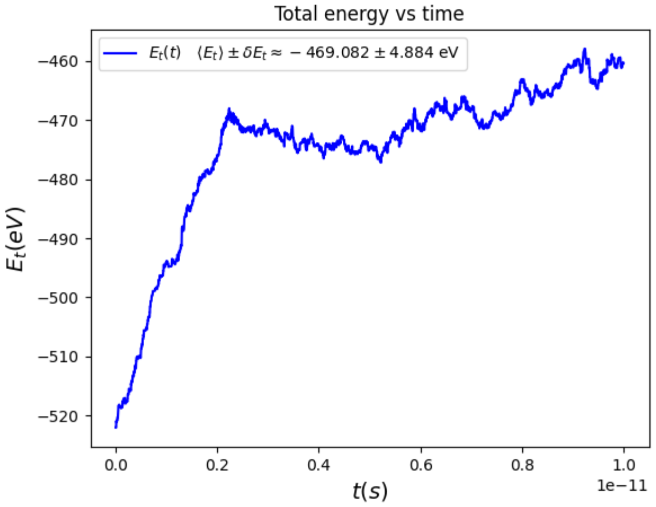


2048

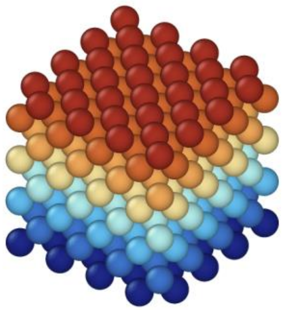
Andersen Thermostat

- **Andersen thermostat** = "thermalizing collisions" between each atom and a virtual bath particle, after which its velocity is reset according to Maxwell-Boltzmann distribution
- Make simulations at higher T in an easier way, because $\langle T \rangle \approx T_{ini}$

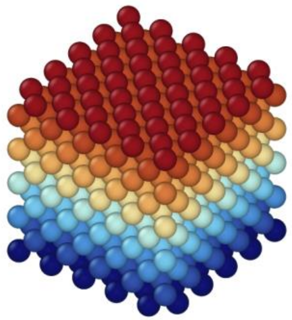




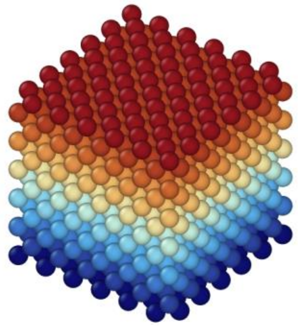
r_V	$r_C + 0.5$ Å
ν	$10^{12}\ s^{-1}$
natoms	256
T	2000 K
dt	2 fs
$t_{tot}\ (ps)$	10



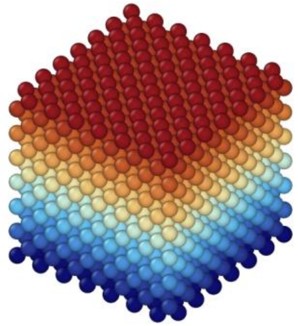
256



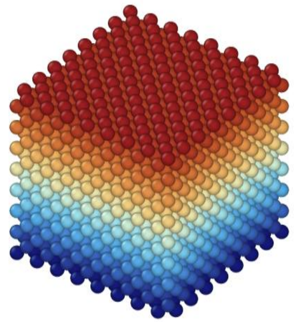
500



864



1372



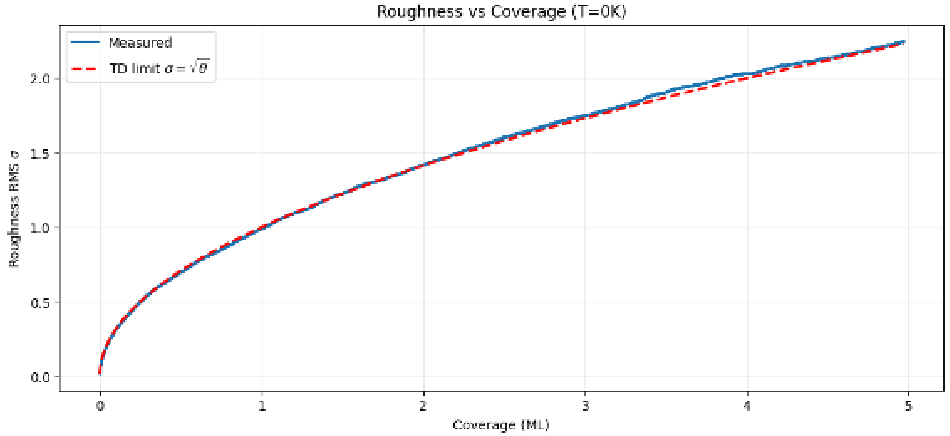
2048

Exercise 2: MC simulations of Ag growth

Part A: Kinetic Monte - Carlo

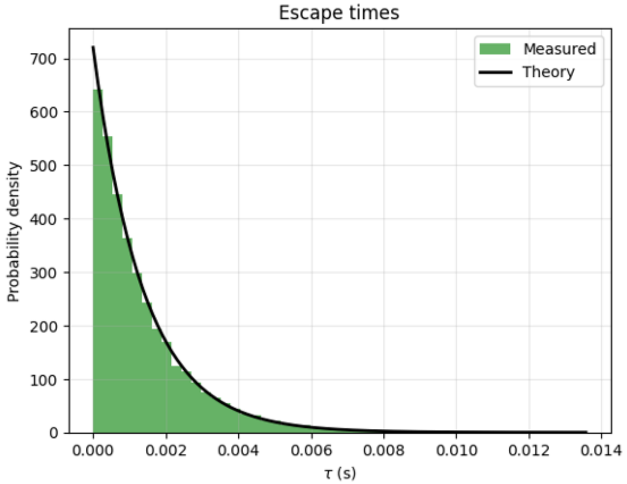
Simulate Ag epitaxy starting from a flat surface at $h(x,y)$ = 0

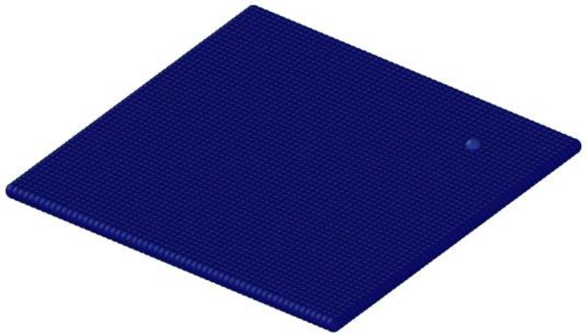
1. KMC – Deposition only



$J_1\ (eV)$	$J_0\ (eV)$	$\nu\ (Hz)$
-0.345	$4J_1$	10^{13}

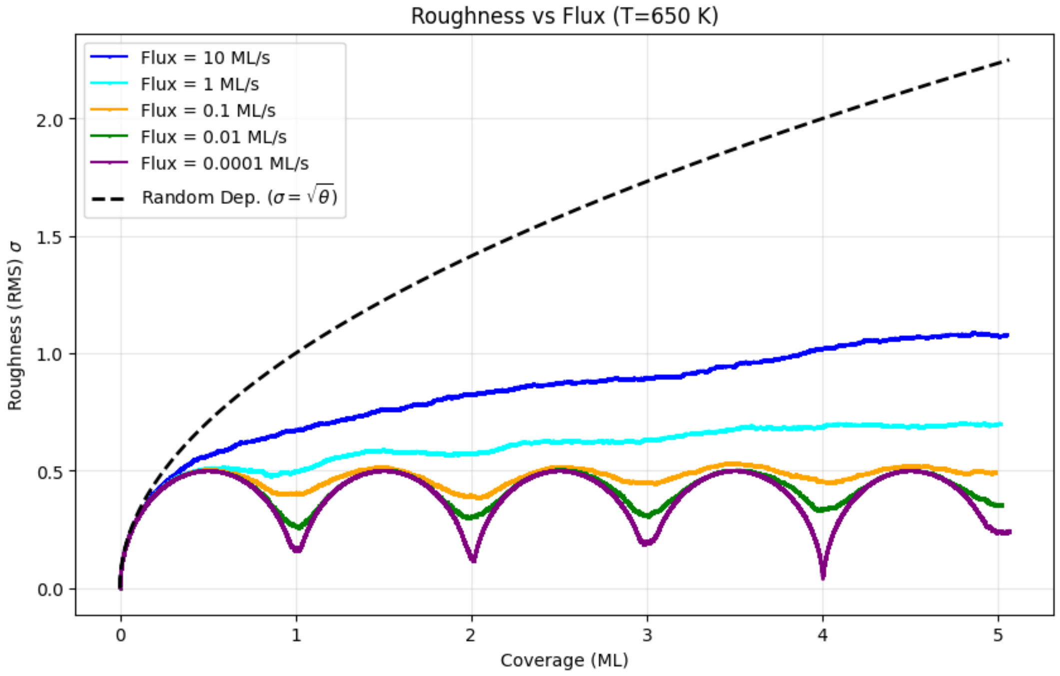
$T\ (K)$	$\phi\ (ML/s)$	$\theta\ (ML)$	$\tau_{mean}\ (s)$	$\langle\tau\rangle - 1/k_{tot}$
0	0.2	5	0.0038	8.71 e-06





- Measured roughness consistent with the square root behaviour at TD limit
- Extracted escape times respect the exponential distribution and their average corresponds to the reciprocal of the deposition rate

2. KMC - Deposition + Diffusion

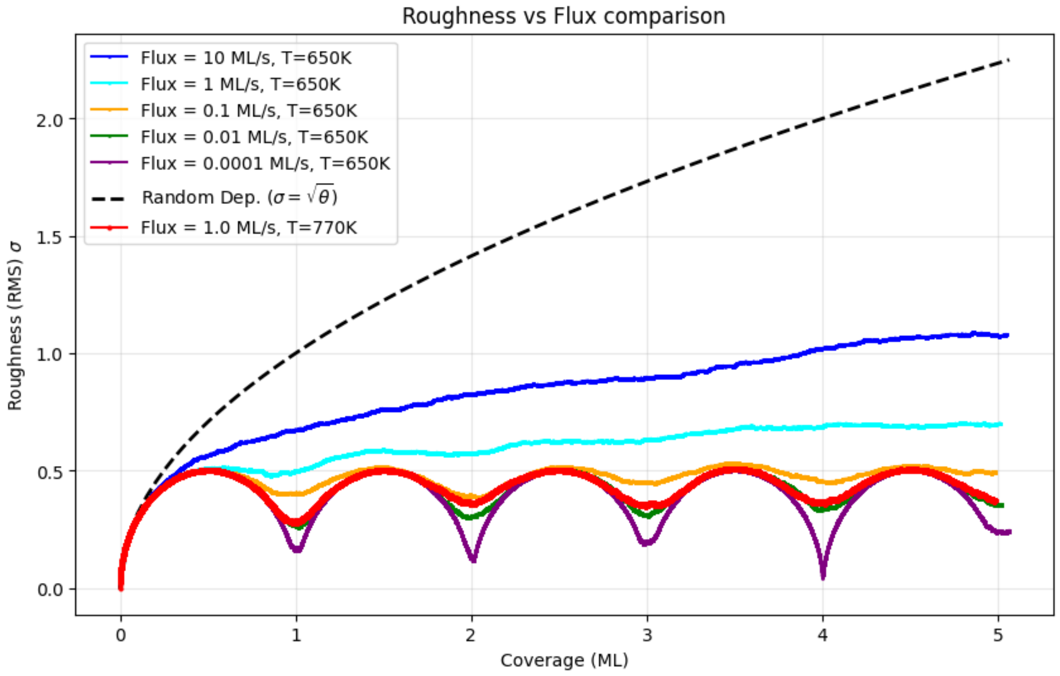


$T\ (K)$	$\phi\ (ML/s)$	$\theta\ (ML)$	$\nu\ (Hz)$	L
650	10, 1, 0.1, 0.01, 0.0001	5	1 e13	60

- 10 ML/s – 1 ML/s: roughness far from random deposition but atoms don't have enough time to diffuse → still disorganized growth

- Layer-by-layer growth: the atoms arrive slowly on the material → they have time to diffuse → almost complete formation of a monolayer before starting the next one → downfall of roughness (smoother surface) corresponding to θ ≈ 1,2,3,4,5

2. KMC - Deposition + Diffusion



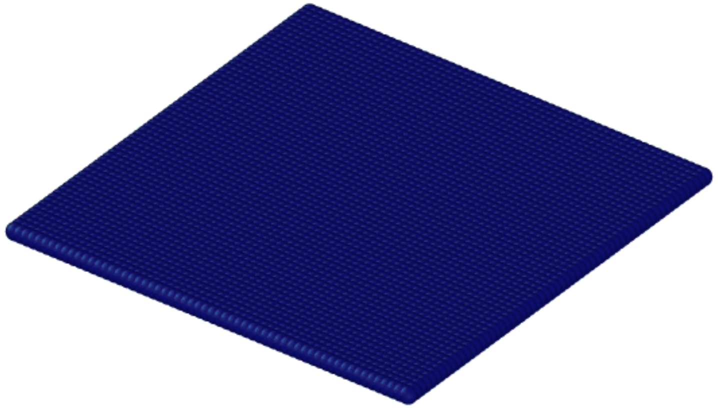
$T\ (K)$	$\phi\ (^{ML}/_{s})$	$\theta\ (ML)$	$\nu\ (Hz)$	L
650	10, 1, 0.1, 0.01, 0.0001	5	1 e13	60

- If the flux is fixed at 1ML/s, can we obtain the same result as 0.01ML/s by tuning the temperature?

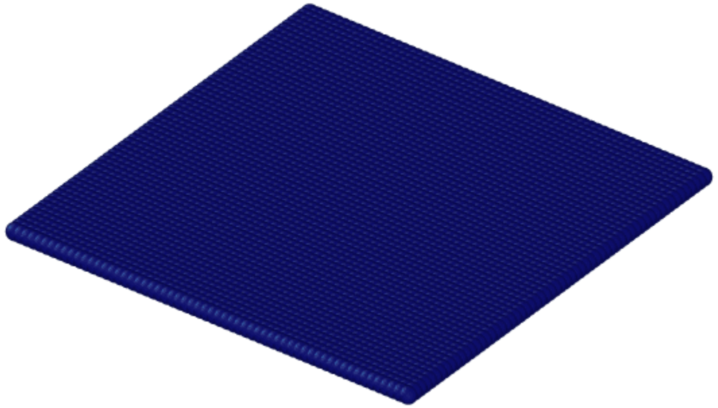
- Yes! We must keep the ratio kdif/flux constant: if the flux gets 100 times bigger (0.01ML/s →1ML/s) so must the diffusion rate → extract T from exponential in kdif → T ≈ 770K

2. KMC - Deposition + Diffusion

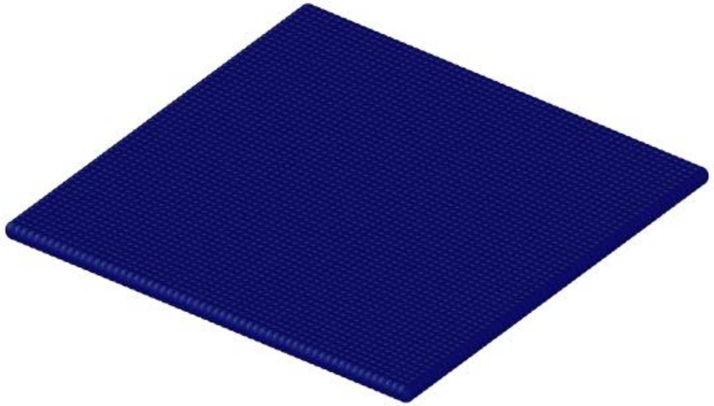
10ML/s



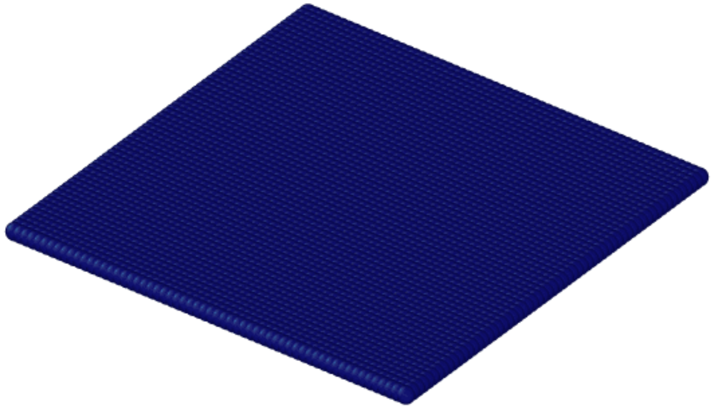
1ML/s



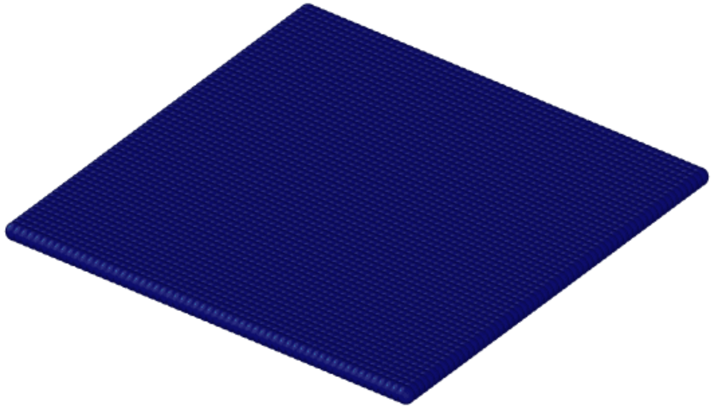
0.01ML/s



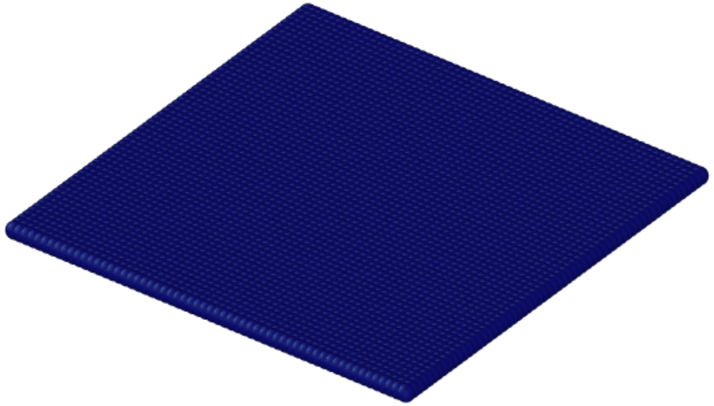
0.1ML/s



0.0001ML/s



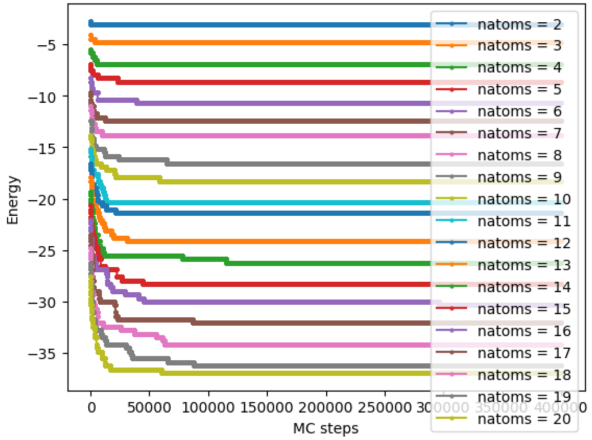
1ML/s (770K)



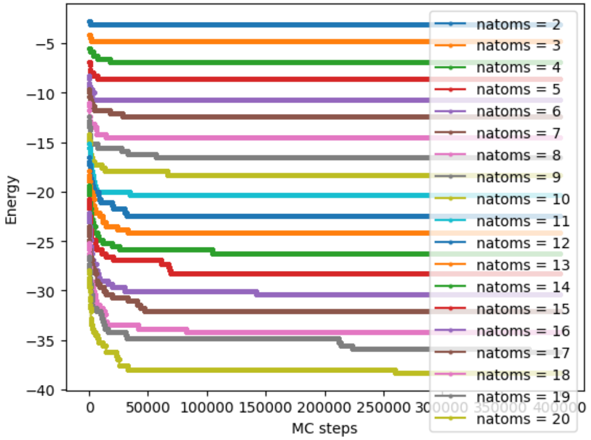
Part B: Metropolis Monte Carlo

Find the equilibrium configuration of a Ag cluster of fixed number of atoms N, randomly deposited on the flat $h(x,y) = 0$ substrate.

1. MMC - Minimum energy and chemical potential



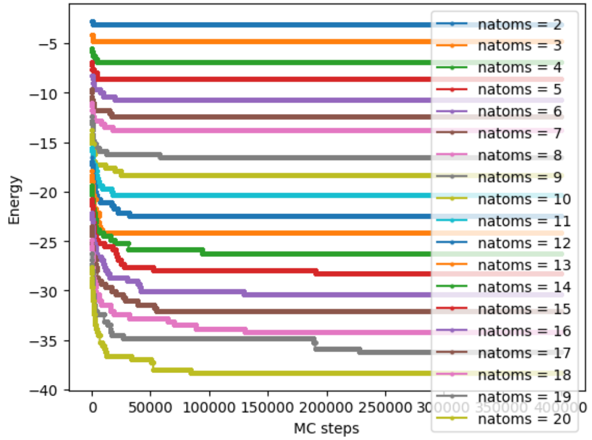
Seed: 16324478



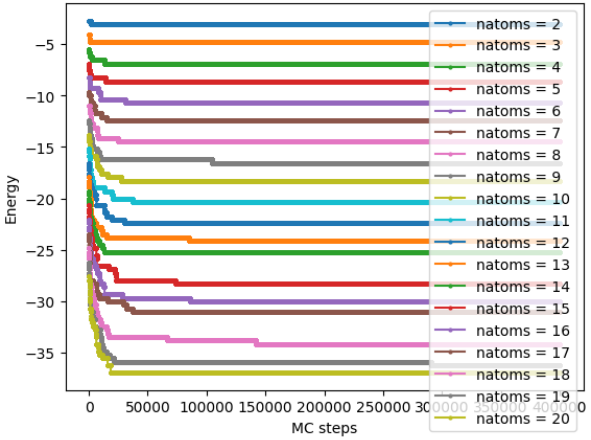
Seed: 42

$T\ (K)$	natoms	L	steps
0	2 to 20	60	1 e6

Seeds	16324478	42	0	1	2026	1926



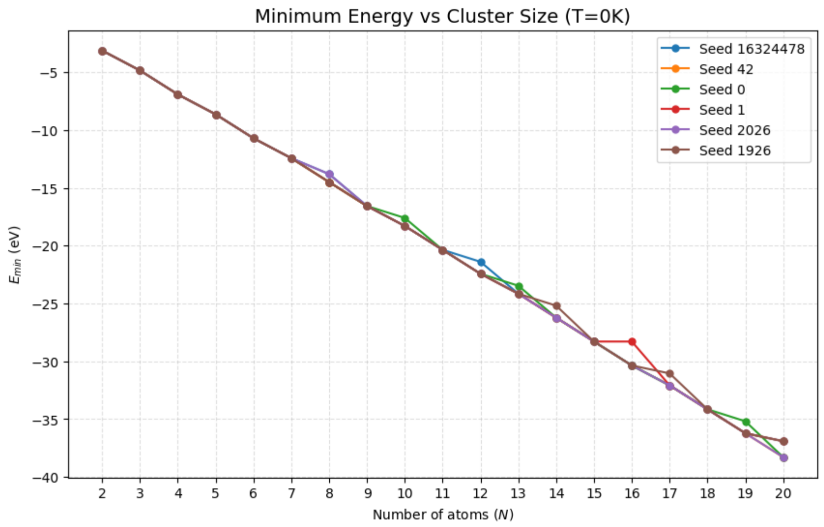
Seed: 2026



Seed: 1926

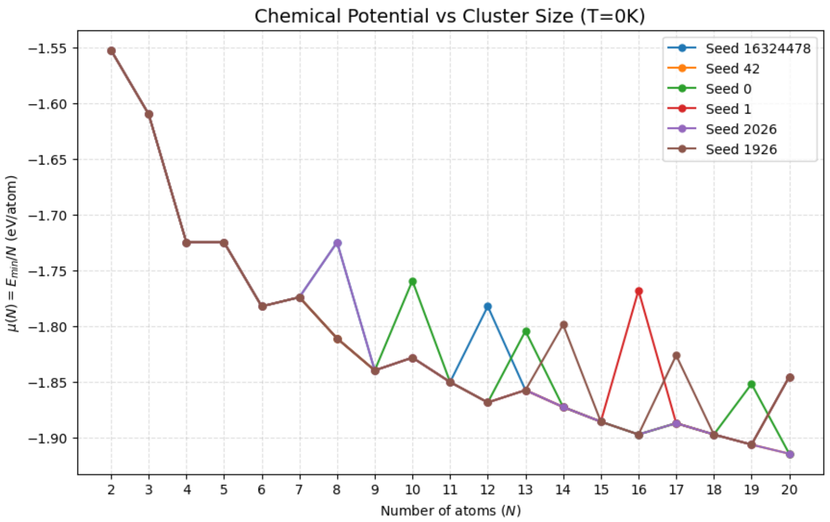
- By running long-enough MC loops we manage to find a global minimum in each case
- We seek to avoid metastable configurations by changing the random generator seed

1. MMC - Minimum energy and chemical potential



$T\ (K)$	natoms	L	steps	neglect
0	2 to 20	60	1 e6	1.5 e5

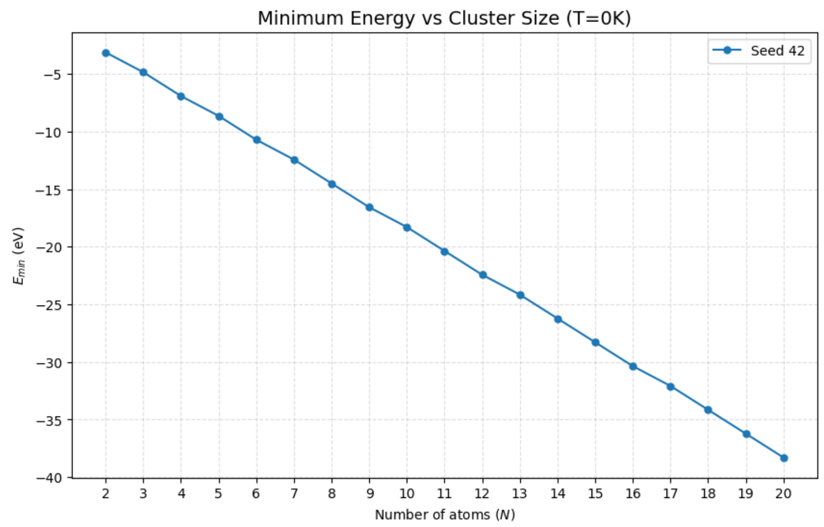
Seeds	16324478	42	0	1	2026	1926

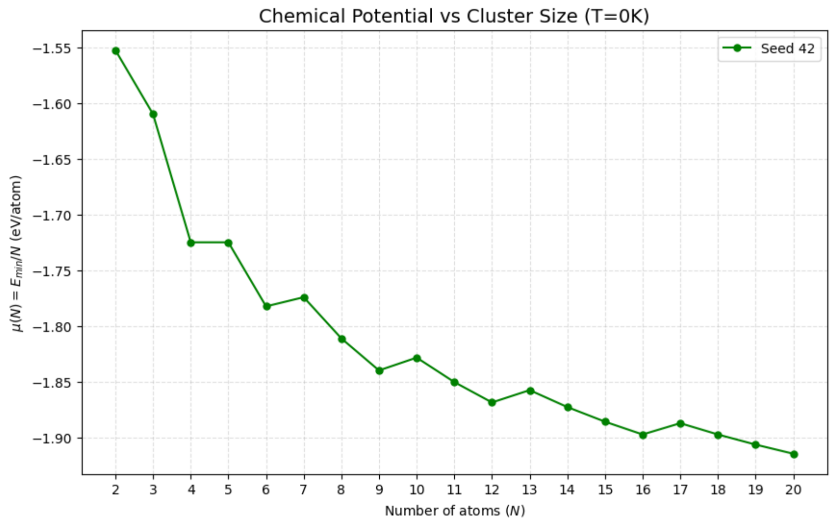


- $E_{minimum}$ monotonously decreasing because we're adding atoms

- Alternating slope of <μ> because it falls when simple geometries are formed (squares or rectangles) and it grows in the next configuration

1. MMC - Minimum energy and chemical potential



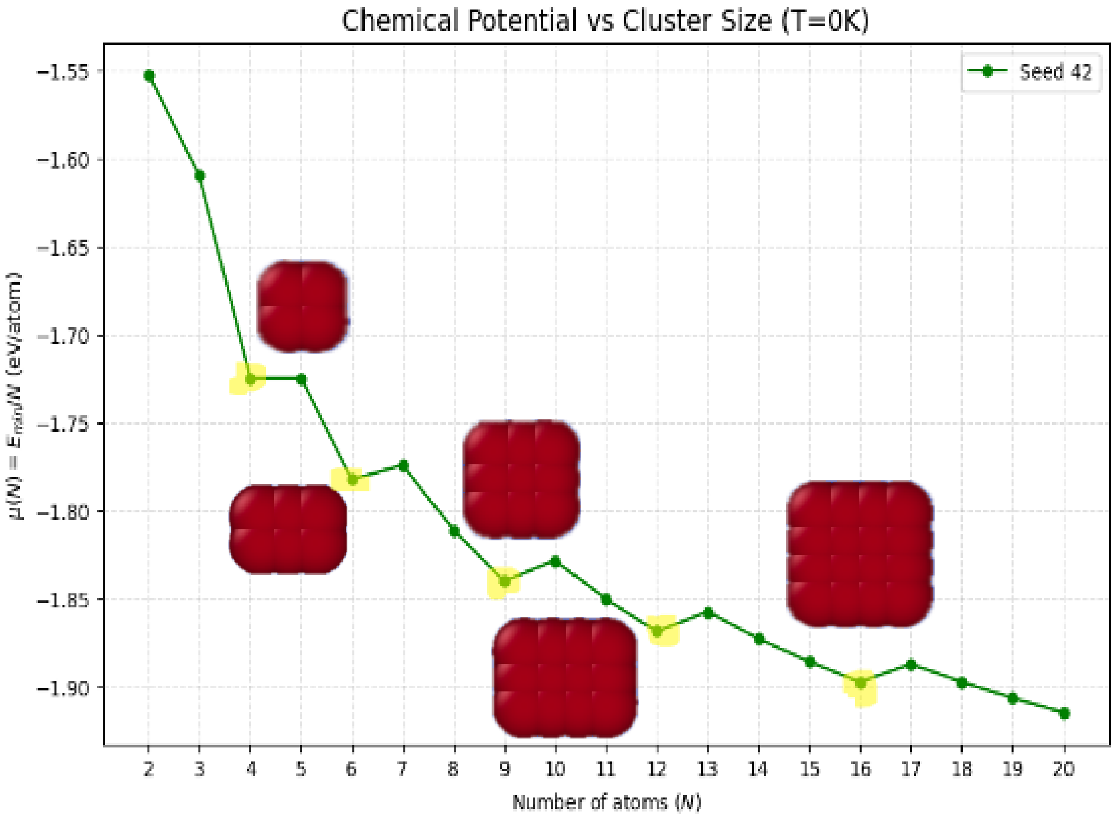


$T\ (K)$	natoms	L	steps	neglect
0	2 to 20	60	1 e6	1.5 e5

Seed	42

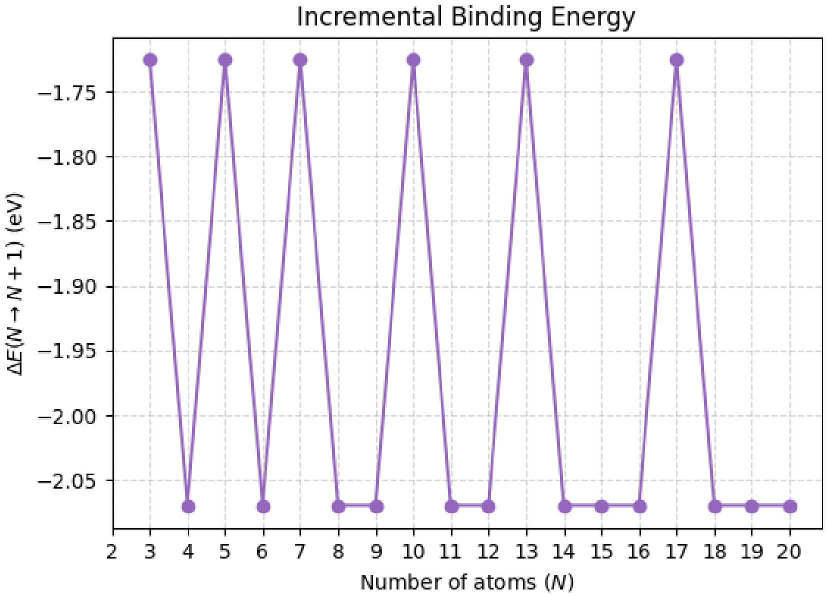
- $E_{minimum}$ monotonously decreasing because we're adding atoms
- Alternating slope of <μ> because it falls when simple geometries are formed (squares or rectangles) and it grows in the next configuration

1. MMC - Minimum energy and chemical potential



$T\ (K)$	natoms	L	steps	neglect
0	2 to 20	60	1 e6	1.5 e5

Seed	42



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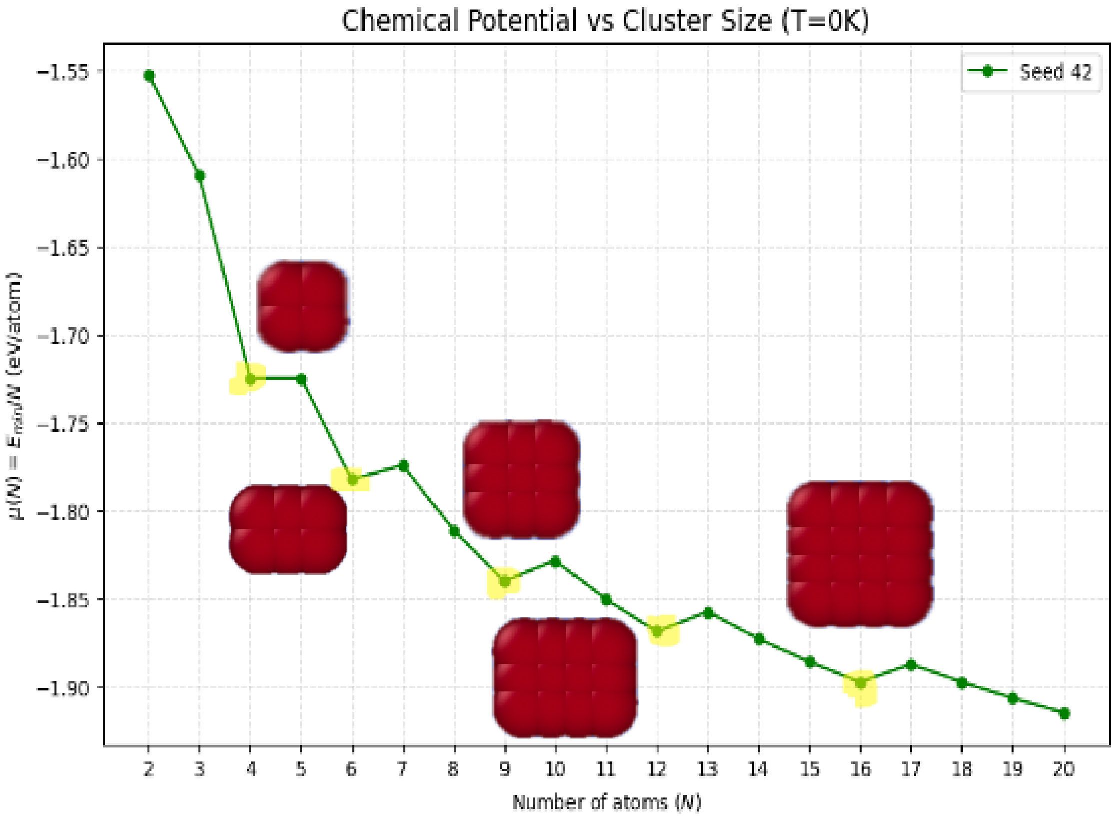
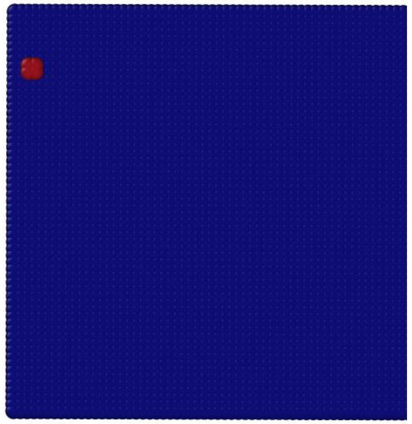
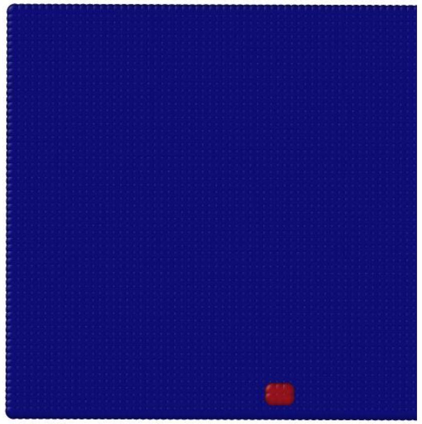
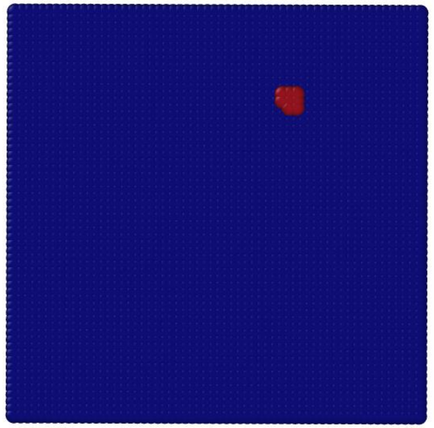




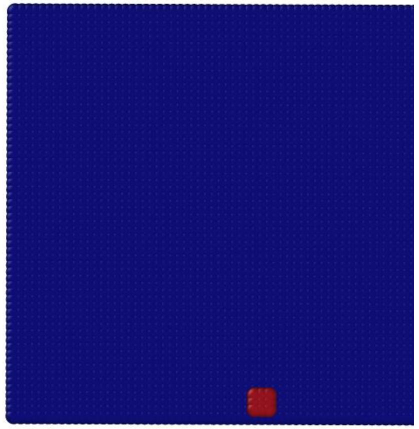
4 atoms



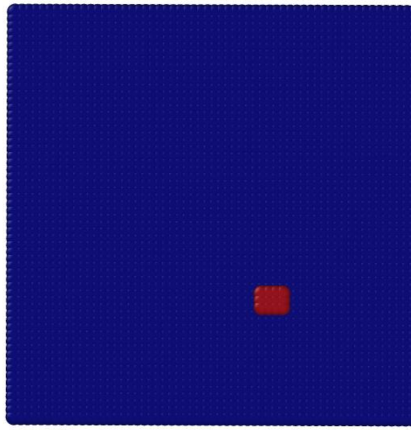
6 atoms



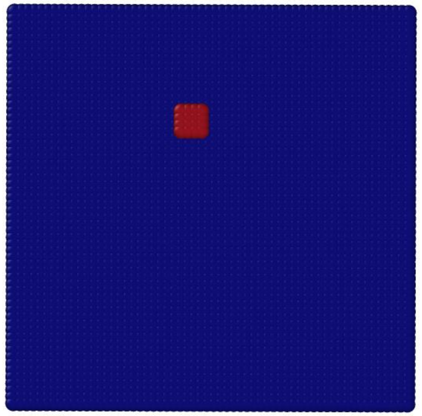
8 atoms



9 atoms

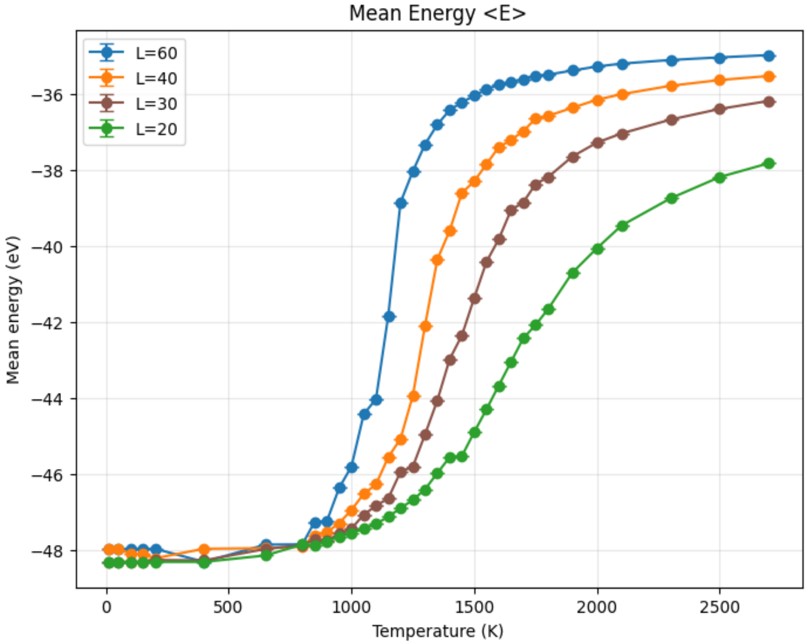


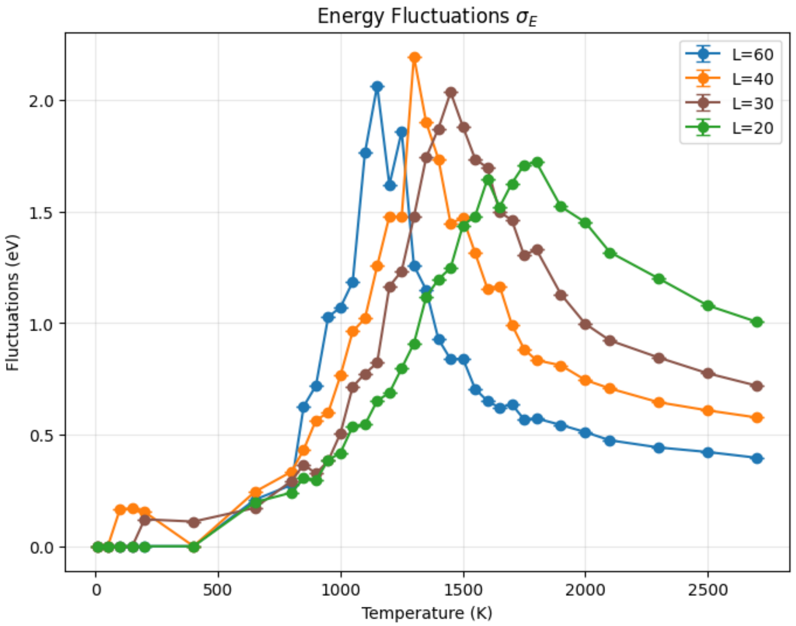
12 atoms



16 atoms

2. MMC – Mean energy and fluctuations



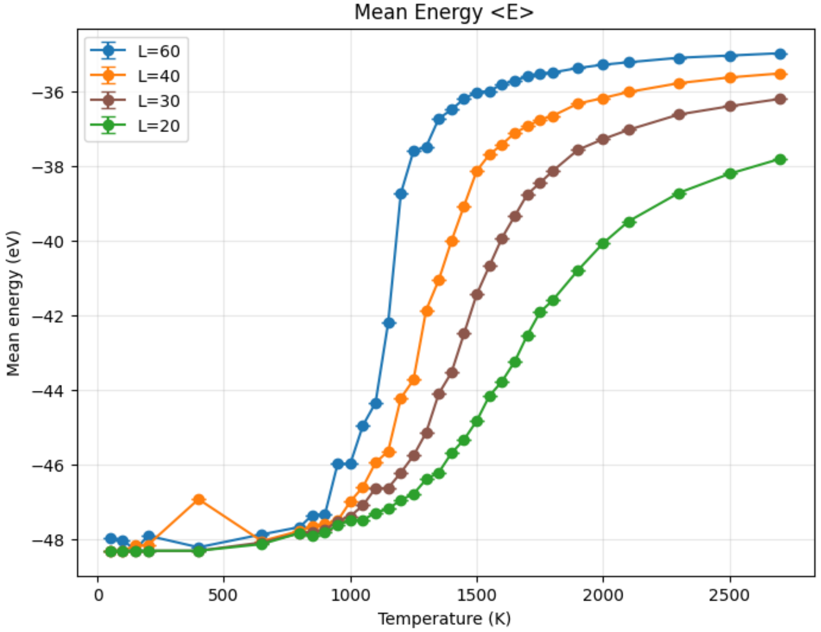


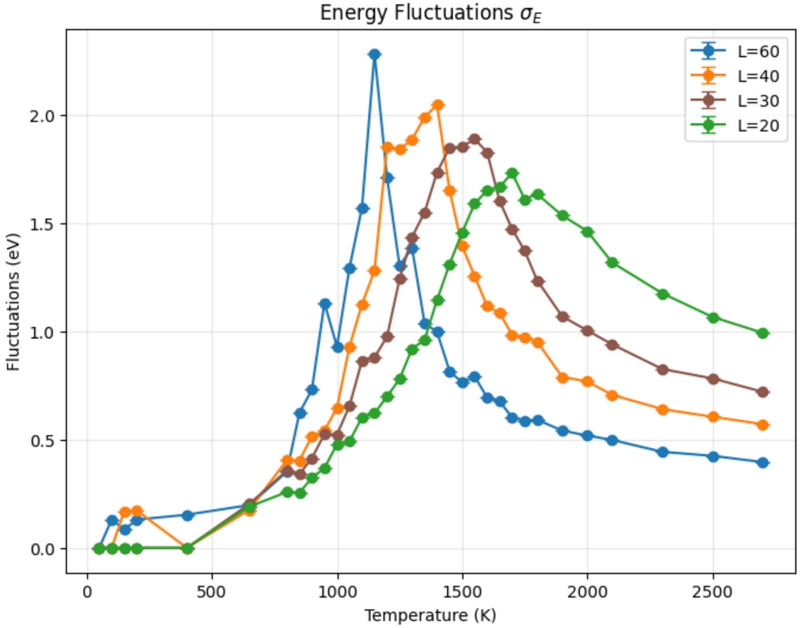
natoms	L	steps	neglect
25	60, 40, 30, 20	1 e6	1.5 e5

Seed	16324478

- Low T: minimum energy and fluctuations → equilibrium reached by froming a compact cluster, which maximizes the number of bonds
- High T: maximum energy and small fluctuations, the temperature is high enough to break the bonds → equilibrium is reached by maximizing entropy.
- Big fluctuations near the inflection point of the sigmoid, where the transition happens
- The bigger the lattice the more configurations are available → entropy maximization favorable at lower T

2. MMC – Mean energy and fluctuations





natoms	L	steps	neglect
25	60, 40, 30, 20	1 e6	1.5 e5

Seed	42

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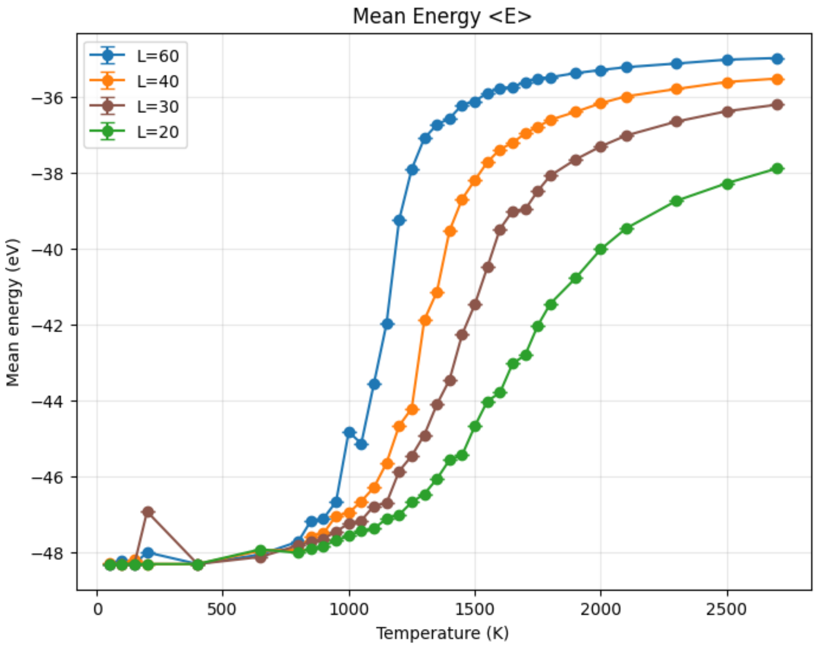

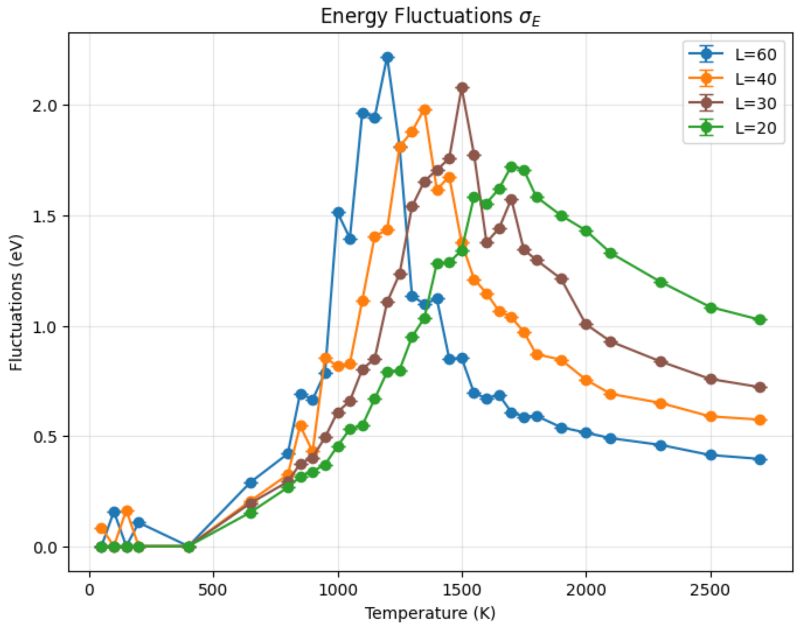

natoms	L	steps	neglect
25	60, 40, 30, 20	1 e6	1.5 e5

Seed	2026

- Low T: minimum energy and fluctuations → equilibrium reached by froming a compact cluster, which maximizes the number of bonds
- High T: maximum energy and small fluctuations, the temperature is high enough to break the bonds → equilibrium is reached by maximizing entropy.
- Big fluctuations near the inflection point of the sigmoid, where the transition happens
- The bigger the lattice the more configurations are available → entropy maximization favorable at lower T

Thank you for your attention